DATA 245 – Machine Learning Technologies
Spring 2026
Group Project Proposal Guidelines

Deadline: Feb 11, 2026

Page Limit: Maximum 3 pages (excluding references)

1. Project Title

Provide a concise and descriptive title that clearly reflects the core idea and objective of your project.

2. Group Members

List all group members (maximum 4 students per group).

There is no need to create groups in Canvas.

3. Dataset

Identify the dataset(s) you will use. Provide a detailed description including:

- Data source

- Size and structure

- Key features or modalities (numerical, categorical, temporal, text, etc.)

- If applicable, include a few example samples

If you plan to collect or generate your own dataset, clearly explain the data generation process and justify realism.

4. Project Summary

Clearly define:

- The problem statement

- Project objectives

- Significance and real-world relevance

- Proposed ML approach or system framework

- Expected impact and contributions

5. Project Background & Literature Review

Provide expanded technical grounding. You must include:

- Review of classical ML methods relevant to your problem

- Prior industry-style systems (Kaggle solutions, applied research papers)

- Discussion of tradeoffs between:

  * Accuracy vs interpretability

  * Bias vs performance

  * Latency vs complexity

At least 15 citations should be included in the proposal.

6. Technical Approach (Mandatory Components)

You must implement at least three model families:

A. Classical Baselines (Required – at least two)

- Logistic / Linear Regression

- k-NN

- Naive Bayes

B. Tree-Based Systems (Required – at least two)

- Decision Trees

- Random Forest

- XGBoost / LightGBM / CatBoost

C. Margin / Kernel Methods (Required – at least one)

- Support Vector Machines (linear and kernel)

Feature Engineering & Data Systems

You must demonstrate:

- Feature transformations

- Handling missing data

- Encoding strategies

- Feature importance analysis

- Data leakage detection

Model Interpretability (at least two required):

- SHAP

- LIME

- Feature importance

- Partial dependence plots

7. Performance Evaluation

Define:

- Evaluation metrics (not just accuracy):

  * ROC-AUC

  * Precision / Recall

  * F1-score

  * Calibration

- Cross-validation strategy

- Error analysis:

  * Which cases does the model fail on?

  * Why do these failures occur?

8. Work Division & Timeline

Specify:

- Each member's technical responsibility

- 3–5 milestones:

  * Data preparation complete

  * Baseline models implemented

  * Final system built

  * Evaluation and report completed

Two progress updates required:

1. In-class presentation

2. Written progress report

9. Project Topics (Open-Ended)

This project is open-ended. Students may propose topics outside the provided list with instructor approval.

10. Submission & Approval Process

- Submit initial proposal

- Receive instructor feedback

- Resubmit if needed

- Begin project after final approval

Grading Emphasis:

Projects will be evaluated based on technical depth, evaluation rigor, novelty, and clarity.

Potential Industry-Driven Project Topics

Real-Time Customer Churn Prediction System
Build a production-style system that predicts which users are likely to churn using behavioral and transactional features, focusing on class imbalance, calibration, and business-driven thresholds.

Credit Risk Scoring & Loan Default Prediction
Design a risk scoring engine for loan approval using structured financial data, emphasizing interpretability, fairness, regulatory constraints, and model governance.

Fraud Detection in Financial Transactions
Detect fraudulent transactions using anomaly detection and supervised classifiers, focusing on rare-event learning, ROC optimization, and cost-sensitive decision rules.

Demand Forecasting for Retail Supply Chains
Predict future product demand using historical sales and temporal features, addressing seasonality, concept drift, and forecasting error decomposition.

Recommender System Using Similarity & Ranking Models
Build a recommendation engine using k-NN and collaborative filtering, with evaluation on ranking metrics, cold-start handling, and business relevance.

Cybersecurity Intrusion Detection System
Develop a machine learning system to detect malicious network activity from tabular logs, optimizing false positive rates and real-time detection constraints.

Healthcare Readmission Risk Prediction
Predict patient readmission risk using clinical tabular data, with strong emphasis on explainability, feature stability, and decision-support reliability.

Marketing Conversion & Customer Propensity Modeling
Model user likelihood to convert on ads or campaigns, focusing on calibration, uplift modeling, and ROI-driven evaluation.

Anomaly Detection in Industrial Sensor Data
Detect abnormal machine behavior using unsupervised and semi-supervised models, emphasizing noise robustness and early fault detection.